

## 29-4a How the Fed Influences the Quantity of Reserves

The first way the Fed can change the money supply is by changing the quantity of reserves. The Fed alters the quantity of reserves either by buying or selling bonds in open-market operations or by making loans to banks (or by some combination of the two). Let's consider each of these methods in turn.

### Open-Market Operations

As we noted earlier, the Fed conducts

[open-market operations \(the purchase and sale of U.S. government bonds by the Fed\)](#)

when it buys or sells government bonds. To increase the money supply, the Fed instructs its bond traders at the New York Fed to buy bonds from the public in the nation's bond markets. By paying for the bonds with newly created dollars, the Fed increases the number of dollars in the economy. Some of these new dollars are held as currency, and some are deposited in banks. Each new dollar held as currency increases the money supply by exactly \$1. Each new dollar deposited in a bank increases the money supply by more than a dollar because it increases reserves and, thereby, increases the amount of money that the banking system can create.

To reduce the money supply, the Fed does just the opposite: It sells government bonds to the public in the nation's bond markets. The public pays for these bonds with its holdings of currency and bank deposits, directly reducing the amount of money in circulation. In addition, as people make withdrawals from banks to buy these bonds from the Fed, banks find themselves with a smaller quantity of reserves. In response, banks reduce the amount of lending, and the process of money creation reverses itself.

Open-market operations are easy to conduct. In fact, the Fed's purchases and sales of government bonds in the nation's bond markets are similar to the transactions that any individual might undertake for her own portfolio. (Of course, when two individuals engage in a purchase or sale with each other, money changes hands, but the amount of money in circulation remains the same.) In addition, the Fed can use open-market operations to change the money supply by a small or large amount on any day without major changes in laws or bank regulations. Therefore, open-market operations are the tool of monetary policy that the Fed uses most often.

### Fed Lending to Banks

The Fed can also increase the quantity of reserves by lending reserves to banks. Banks borrow from the Fed when they feel they do not have enough reserves on hand, either to satisfy bank regulators, meet depositor withdrawals, make new loans, or for some other business reason.

There are various ways banks can borrow from the Fed. Traditionally, banks borrow from the Fed's *discount window* and pay an interest rate on that loan called the discount rate (the interest rate on the loans that the Fed makes to banks). When the Fed makes such a loan to a bank, the banking system has more reserves than it otherwise would, and these additional reserves allow the banking system to create more money.

The Fed can alter the money supply by changing the discount rate. A higher discount rate discourages banks from borrowing from the Fed, decreasing the quantity of reserves in the banking system and in turn the money supply. Conversely, a lower discount rate encourages banks to borrow from the Fed, increasing the quantity of reserves and the money supply.

At times, the Fed has set up other mechanisms for banks to borrow from it. For example, from 2007 to 2010, under the *Term Auction Facility*, the Fed set a quantity of funds it wanted to lend to banks, and eligible banks then bid to borrow those funds. The loans went to the highest eligible bidders—that is, to the banks that had acceptable collateral and offered to pay the highest interest rate. Unlike at the discount window, where the Fed sets the price of a loan and the banks determine the quantity of borrowing, at the Term Auction Facility the Fed set the quantity of borrowing and competitive bidding among banks determined the price. The more funds the Fed made available, the greater the quantity of reserves and the larger the money supply.

The Fed lends to banks not only to control the money supply but also to help financial institutions when they are in trouble. For example, when the stock market crashed by **22** percent on October 19, 1987, many Wall Street brokerage firms found themselves temporarily in need of funds to finance the high volume of stock trading. The next morning, before the stock market opened, Fed Chair Alan Greenspan announced the Fed's "readiness to serve as a source of liquidity to support the economic and financial system." Many economists believe that Greenspan's reaction to the stock crash was an important reason it had few repercussions.

Similarly, in 2008 and 2009, a fall in housing prices throughout the United States led to a sharp rise in the number of homeowners defaulting on their mortgage loans, and many financial institutions holding those mortgages ran into trouble. In an attempt to prevent these events from having broader economic ramifications, the Fed provided many billions of dollars in loans to financial institutions in distress.